



UNIVERSITY COLLEGE OF ENGINEERING SCIENCE AND TECHNOLOGY HYDERABAD
JAWAHARLAL NEHRU TECHNOLOGICAL UNIVERSITY HYDERABAD
 Kukatpally, Hyderabad - 500 085, Telangana State, INDIA

MID EXAMINATION ANSWER BOOKLET

H.T. No.

2	3	0	1	1	M	3	1	1	5
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NAME OF THE EXAMINATION B.Tech Third Year SEM- I, MIP- I Examination

SUBJECT DRMS

BRANCH/SPECIALIZATION CSE - JPPMP

DATE OF EXAM 06-03-2025

[Signature]
Signature of Invigilator

Please tick (✓) the questions and bits answered by you
in the space provided above the dotted line

	1	2	3	4	5	6	7	8	9	10
Part A										

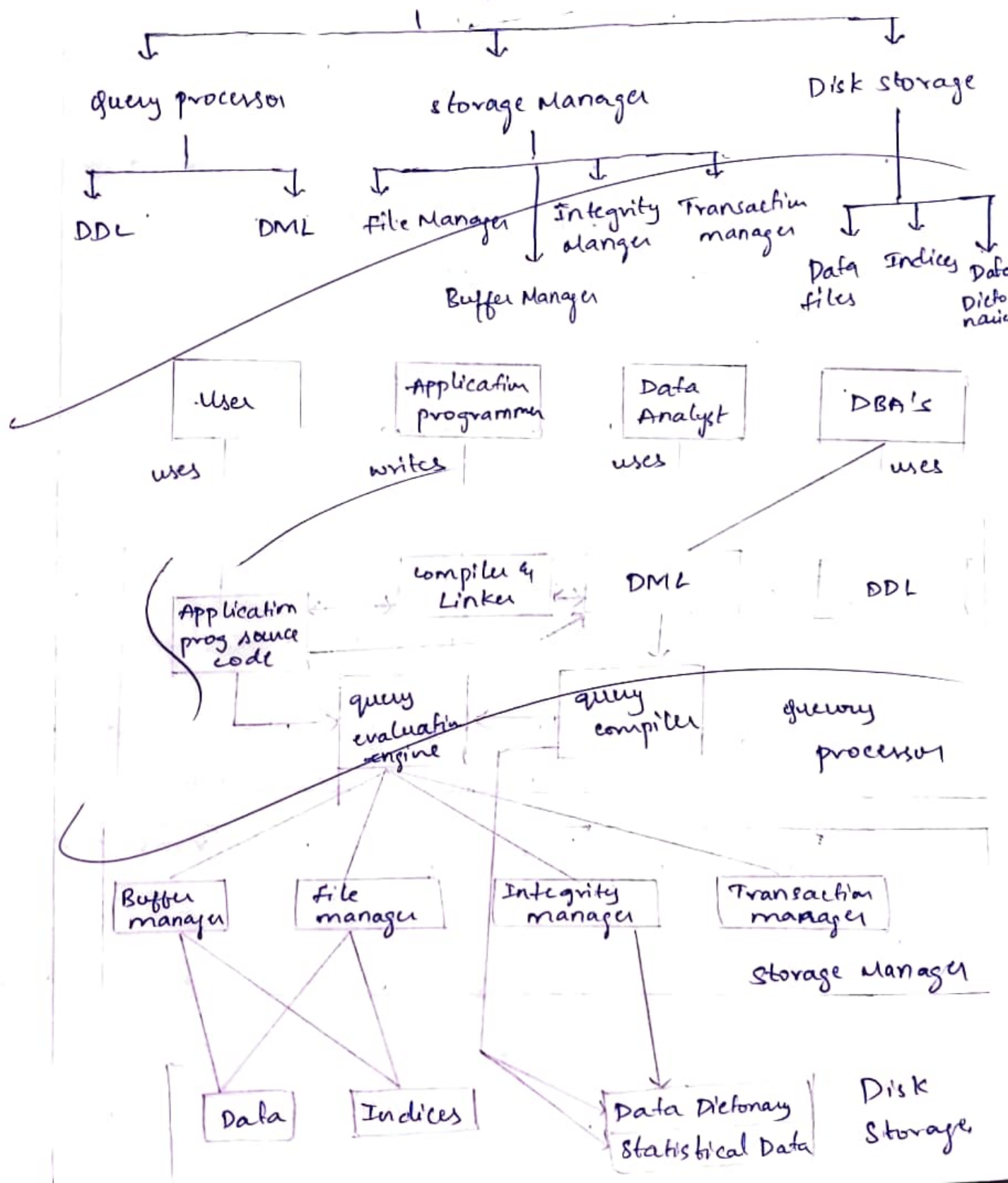
Part B

Question No.	1	2	3	4	5	6	7	8
Tick Marks		✓	✓	✓	✓			
Tick Marks								
Tick Marks								
Question Wise Total Marks								
(Part A + Part B) = Grand Total (In Figures)								
In words								

INSTRUCTIONS TO CANDIDATES OVERLEAF

- 5) DBMS : Data Base Management System, is a software to access the database in form of queries such as creating table, insertion, deletion, data manipulations etc.

structure of DBMS



Query processor is divided into:

- Data Definition: It defines the data, such as creating etc.
- Data manipulation:

The Data is manipulated using DML in the DataBase performs of operation such as select, where clause, Alter, Drop etc,

- DML compiler: The ~~processed~~ query raised by the user is compiled by DML compiler and gives a certain output.

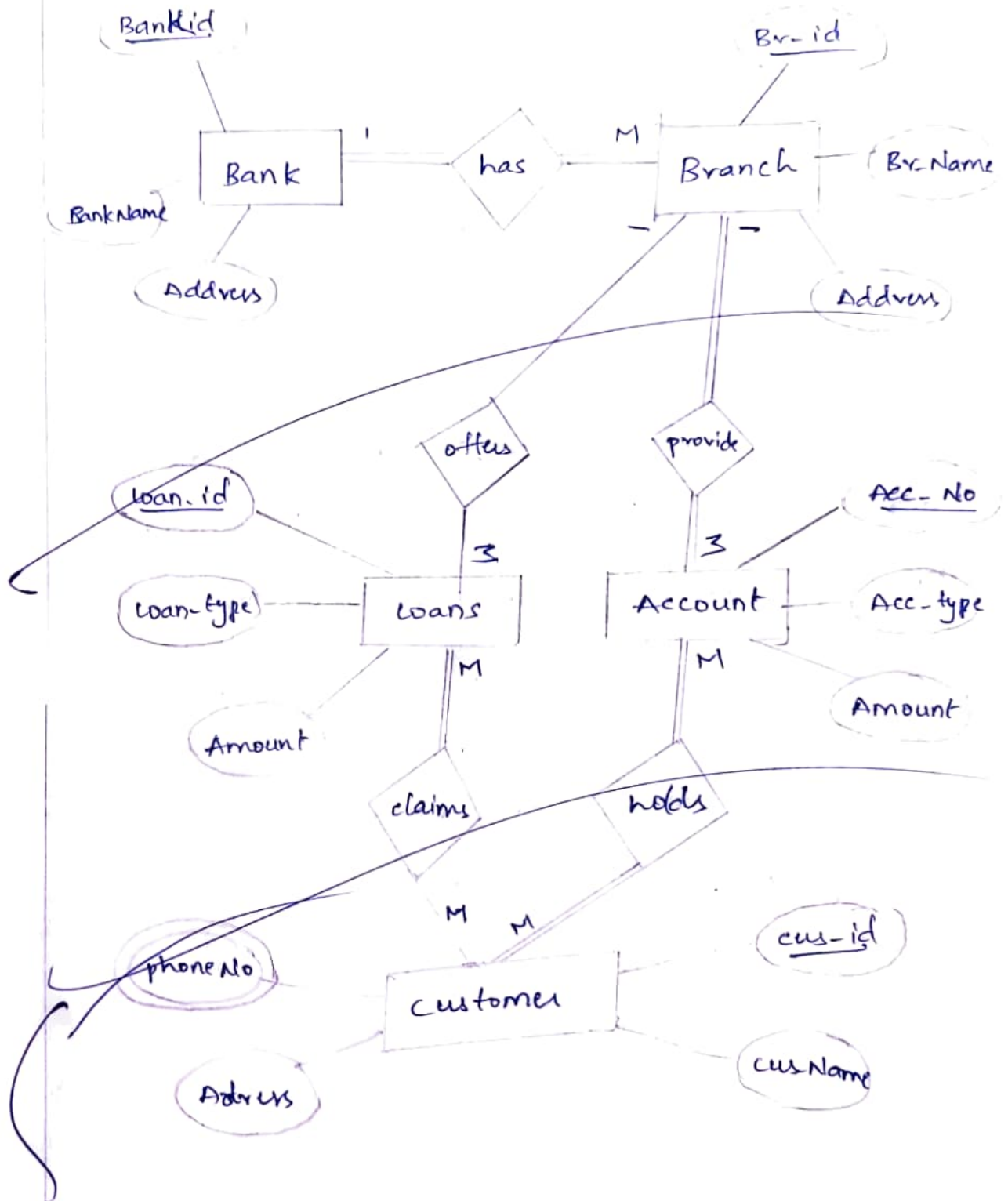
Storage Manager:

- File Manager: performs operation related to files such as creating file, read, write etc,
- Buffer manager: process the info is granted into main memory using cache
- Integrity manager: Provides integrity of Database following certain rules.
- Transaction manager: manages complex transactions.

Disk storage:

- Data: How the Data is stored as what data is stored
- Indices: performs operations on Data.
- Data dictionaries: provides statistical data of certain data entered.

3) ER Model for Banking enterprise.



In Banking Enterprises:

Entities are Bank, Branch, Loan, account, customer

Attributes of Bank are

Bank-id, Bank-name, Address

Attributes of Branch are Br-id, Br-name, Address

which are represented by ellipse.

The relations between ~~the~~ entities are represented by Diamond.

The relation type is also represented as

Many to many between bank & Branch,

~~Many~~ customer and Account, customer and Loan

one to many between Branch & account,

Branch and Loan.

As one Branch contains many accounts and can offer many loans.

4) Given,

Sailors (sailor-id, sailor-name, Rating, Age)

Reserves (sailor-id, Boat-id, Day)

Boats (Boat-id, Boat-name, color)

(i) Name of sailor who have reserved atleast one boat

select S.sailor-name from sailor S

JOIN reserves R ON S.sailor-id = R.sailor-id;

O/P: Gives sailors name who reserved atleast one boat.

(ii) Names of sailors who have reserved boat No. 120

select S.sailor-name from sailor S

JOIN reserves R ON S.sailor-id = R.sailor-id

where Boat-id = 120 ;

Sample O/P:

sailor-name
Ajay

(iii) Find the Average age of sailor with age > 20

select AVG(age) as avg-age from sailor
where age > 20 ;

O/P:

avg-age
23

(iv) Rating of sailor who reserved red boats.

Select s.sailor_id, s.sailor_name, s.rating
from sailor s

JOIN reserves R ON s.sailor_id = R.sailor_id

JOIN Boat B ON R.Boat_id = B.Boat_id

where B.color = 'red';

O/p:

sailor_id	sailor_name	Rating
20	Max	7
22	Henry	8

(v) Number of boats reserved by each sailor.

Select ~~sailor~~ s.sailor_id, COUNT(sailor_id) as
num-boats-reserved

JOIN reserves R ON s.sailor_id = R.sailor_id
GROUP BY s.sailor_name;

O/p:

sailor_name	num-boats-reserved
Max	5
Henry	2
Ajay	3

- 2) Fundamental operation of relational algebra.
- (1) select (σ)
 - (2) project (π)
 - (3) Union ($\cup$)
 - (4) Set Difference ($-$)
 - (5) Cross product ($\times$)
 - (6) Rename.

select (σ) :

selects the attributes of given condition it may be single or more than one

$\sigma_{\langle \text{condition} \rangle} (\text{table-name})$

It is denoted by σ (sigma).

Eg:

$\sigma_{\text{customer-id, cus-name}} (\text{customer})$

customer-id	cus-name
1	X

project (π) :

Gives only a certain row (tuple) from the table including all attribute values.

$\pi_{\langle \text{condition} \rangle} (\text{table-name})$

Denoted by π (π).

Eg: $\pi_{\text{cus_name}}$ (customer)

cus_name
X
Y
Z

Union ($\cup$):

It perform same as union function in set theory. It combines two rows (tuple) of given conditions & includes all attributes

It is denoted by $\cup$. Notation $R \cup S$

Eg: To get Display student id with marks.

$\pi_{\text{std-id}}(\text{student}) \cup \pi_{\text{marks}}(\text{education})$

std-id	marks
X	70
Y	85
Z	88

Set Difference ($-$):

Performs operation by of attributes of table with which are not in another table present

It is denoted by $-$

Notation: $R - S$

Defined as: $\{t \mid t \in R, \text{ and } t \notin S\}$

Cross product:

Combines two different tables but it should contain same attributes and same Domain types

No. of

It is denoted by $\times$.

Notation: $R \times S$

Eg: R table

sid	Grade
1	A
2	B

S Table

sid	Grade
3	B
4	C

$R \times S \Rightarrow$

sid	Grade	sid	Grade
1	A	3	B
2	B	4	C

Rename (ρ):

used to rename a certain attribute

It is denoted by ' ρ ' :